Md. Taifur Hossain.

SOFTWARE ENGINEERING · COMPUTER VISION · ML

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Final-year Computer Science undergraduate who ships working software. Built and published **six projects on GitHub** across 3D vision, model compression, developer tooling, and applied AI, four with tagged releases and three with tests and CI, plus a **first-author paper accepted at ICCA 2026 (ACM)**. They range from a Next.js learning platform to a hybrid medical expert system. Comfortable across C, C++, C#, Python, and JavaScript, and used to owning a problem from prototype through to a documented release. Open to software engineering internships across backend, full-stack, systems, and ML.

TECHNICAL SKILLS

LANGUAGES	C, C++, C#, Python, JavaScript / TypeScript, SQL, Bash
WEB & FRAMEWORKS	Next.js, React, Node.js, Tailwind CSS, Flask, Streamlit, Three.js, Vite, Prisma, SQLite, HTML / CSS
MACHINE LEARNING	PyTorch, scikit-learn, NumPy, RAG and embeddings, Bayesian inference, heuristic search (A*), model quantization
3D & VISION	3D Gaussian Splatting, NeRF, Nerfstudio, gsplat, COLMAP
TOOLS & PRACTICE	Git, Docker, GitHub Actions (CI/CD), GHCR, Linux, pytest, ruff, ESLint, Agile / Scrum

PROJECTS

Lumina, AI Learning Management System

GitHub

AI-assisted learning platform built across Agile sprints, with a retrieval tutor grounded in real course material.

- Full-stack Next.js 16 and React 19 application in TypeScript, roughly **7,500 lines**, backed by Prisma and SQLite.
- Retrieval-augmented tutor that answers from uploaded course materials and shows the sources it used.
- Role-based access for Student, Instructor, and Admin, with JWT sessions, bcrypt password hashing, and Zod input validation.
- Adaptive learning paths, gamification (XP, badges, streaks), and instructor analytics dashboards.
- Runs fully offline against a mock AI provider, or against Google Gemini by setting a single environment variable.

Tech Stack: Next.js, React, TypeScript, Tailwind CSS, Prisma, SQLite, Google Gemini, Zod, Recharts.

splat-slim

GitHub

Training-free compression tool for trained 3D Gaussian Splatting models.

- Compresses trained 3DGS models by **roughly 74% with under 3 dB quality loss**, with no retraining.
- Four-stage pipeline: adaptive opacity pruning, outlier clipping, spherical-harmonic reduction, and quantization.
- Three quantization modes: FP16, mixed FP16/INT8, and Morton-ordered sub-group INT8, the only one that writes per-group ranges to a sidecar `.meta.npz`. Unquantized output opens in standard viewers; quantized PLYs need a range-aware reader.
- Shipped as a CLI and a library, with automated tests and linting on every push and tagged releases.

Tech Stack: Python, Typer, NumPy, plyfile, pytest, ruff, GitHub Actions.

AI-Based Medical Expert System

GitHub

Hybrid expert system combining five AI techniques into a single diagnosis pipeline.

- Combines rule-based reasoning with confidence factors, **A* search**, Best-First search, **Bayesian inference**, and a Decision Tree plus MLP neural network.
- Merges all five outputs through a weighted ensemble to produce a ranked diagnosis with confidence scores.
- Encodes 10 IF-THEN rules over 15 binary symptom features across 6 conditions, and benchmarks the pipeline against a real dataset.
- Serves the whole system through a Flask web interface with a symptom checklist and a results dashboard.

Tech Stack: Python, Flask, scikit-learn, NumPy.

nerf-prep GitHub

One-command Dockerized pipeline turning a folder of photos into a training-ready 3D dataset.

- Packaged a fragile multi-tool chain (COLMAP to Nerfstudio) into **a single reproducible Docker command**, published to GHCR automatically by CI.
- Automatic matcher selection by capture type, plus GPU and CPU auto-detection with a fallback, so the same command runs on a cloud GPU or a laptop.
- Validates input before a long run, flagging double extensions and too-few images, then prints a registration-rate summary. Verified on a real 40-photo capture at **40 of 40 registered**.

Tech Stack: Python, Typer, Docker, COLMAP, Nerfstudio, GitHub Actions, GHCR.

paper-chat GitHub

Retrieval-augmented question answering over a PDF library, with citations you can verify.

- **Every answer cites the exact paper and page** it came from, and the app declines to answer when the sources do not cover the question.
- Parses PDFs per page so each retrieved chunk carries a (paper, page) tag, enabling cross-paper comparison questions.
- Swappable LLM backend (Groq or Gemini) behind a small interface, with retrieval and citation logic unit-tested using fakes, so tests run with no API key.

Tech Stack: Python, Streamlit, sentence-transformers, PyMuPDF, NumPy.

splat-lab Live Demo · GitHub

Browser-based, real-time demo of the compression pipeline.

- Load a 3D scene and watch file size and visual quality trade off **live on the GPU** while dragging sliders.
- Ported the Python compression maths to TypeScript so the browser reproduces the same results, running entirely client-side with no backend.

Tech Stack: TypeScript, Three.js, Vite, GitHub Pages.

OPEN SOURCE CONTRIBUTIONS

- **gsplat PR #1063** (open): opacity × area weighted K-means for the `PngCompression` SH codebook, **+0.111 dB** mean PSNR on 9 MipNeRF360 scenes and +0.052 dB on Tanks & Temples, better on every scene at the same size and format.
- **gsplat PR #1061** (open): fixed `PngCompression` failing on `sh_degree=0` scenes and empty tensors.

PUBLICATIONS

Beyond Naive INT8: Adaptive Post-Training Compression for 3D Gaussian Splatting · ACCEPTED

M. T. Hossain, S. S. Sourav, M. A. Hossain. Proc. ICCA 2026 (ACM), Dhaka.

- First author. Built the full experimental pipeline and evaluated it across three scenes on a single NVIDIA T4 (Kaggle).
- Achieved a **74% size reduction without retraining**, and diagnosed a quantization failure mode that a sub-group range scheme recovers to within 0.1 to 0.7 dB.

Rendering-aware SH vector quantization for compact 3DGS: a pre-registered replication and dissection · IN PREPARATION

- Pre-registered study run from a gsplat fork, testing whether a rendering-derived codebook metric holds on held-out test views, with controlled fine-tuning ablations.

EDUCATION

International University of Business Agriculture & Technology (IUBAT) Dhaka, BD

B.Sc. in Computer Science & Engineering (BCSE) · CGPA 3.3 / 4.00 Expected 2027

Higher Secondary Certificate (HSC) · GPA 5.00 / 5.00 Dhaka, BD · 2019 - 2021

Secondary School Certificate (SSC) · GPA 5.00 / 5.00 Dhaka, BD · 2018

LANGUAGES

ENGLISH	Proficient
BANGLA	Native

REFERENCES

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